STA2001 Probability and Statistics I

Assignment 3

Deadline: 23:59 pm, Sep 26th, Friday

1. For each of the following determine the constant c so that $f(x)$ satisfies the conditions of being a pmf for a random variable X:
   (a) $f(x) = x/c$, $x = 1, 2, 3, 4$. $c=10$
   (b) $f(x) = c(1/4)^x$, $x = 1, 2, 3, \cdots$ $c=3$
   (c) $f(x) = \frac{c}{(x+1)(x+2)}$, $x = 0, 1, 2, 3, \cdots$ Hint: Write $f(x) = c/(x+1) - c/(x+2)$.
   $c=1$

2. Let X be the number of accidents per week in a factory. Let the pmf of X be

$$f(x) = \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}, \quad x = 0, 1, 2, \cdots$$

Find the conditional probability of $X \geq 4$, given that $X \geq 1$.

3. Find $\mathbb{E}[X]$ for each of the distributions given in Q1.

(b.)&(c.) 结合级数知识考察.

4. Let X be a discrete random variable whose value is in $\{0, 1, 2, \cdots\}$. Prove that

<Proof>.

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} \mathbb{P}(X > k).$$

Hint: Let

$$1_{\{x>k\}} = \begin{cases} 1 & \text{for } x > k \\ 0 & \text{otherwise.} \end{cases}$$

We have $\mathbb{P}(X > k) = \sum_{x=0}^{\infty} 1_{\{x>k\}} \mathbb{P}(X = x)$. We also assume that $\sum_{x=0}^{\infty} \sum_{y=0}^{\infty} g(x, y) = \sum_{y=0}^{\infty} \sum_{x=0}^{\infty} g(x, y)$ is valid for non-negative multi-variable function $g(x, y)$.

5. Let the pmf of X be defined by $f(x) = 6/(\pi^2 x^2)$, $x = 1, 2, 3, \cdots$. Show that $\mathbb{E}[X] = +\infty$ and thus, does not exist.

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T1. (a.) $\frac{1+2+3+4}{c} = \frac{10}{c} = 1 \Rightarrow c=10$

(b.) $\sum_{x=1}^{\infty} c\cdot(\frac{1}{4})^x = c\cdot(\frac{\frac{1}{4}}{1-\frac{1}{4}}) = \frac{c}{3} = 1 \Rightarrow c=3$

(c.) $\sum_{x=0}^{\infty} c\cdot\frac{1}{(x+1)(x+2)} = c\cdot\sum_{x=0}^{\infty}(\frac{1}{x+1}-\frac{1}{x+2}) = c\cdot(1-\lim_{x\to\infty}\frac{1}{x+2}) = c = 1 \Rightarrow c=1.$

T2. $P(X\geq 4 | X\geq 1) = \frac{P(X\geq 4)}{P(X\geq 1)}$. $P(X\geq 1) = 1-f_X(0) = 1-(1-\frac{1}{2}) = \frac{1}{2}$ $P(X\geq 4) = 1-\sum_{x=0}^{3} f(x) = 1-(1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\frac{1}{4}-\frac{1}{5}) = \frac{1}{5}.$

$\therefore P(X\geq 4 | X\geq 1) = \frac{\frac{1}{5}}{\frac{1}{2}} = \frac{2}{5}.$

T3. (a.) $E(X) = \sum_{x=1}^{4} x f(x) = 1\times\frac{1}{10}+2\times\frac{2}{10}+3\times\frac{3}{10}+4\times\frac{4}{10} = \frac{30}{10} = 3$

(b.) $E(X) = \underbrace{3\cdot\sum_{x=1}^{\infty} x\cdot(\frac{1}{4})^x}_{①} = \frac{3\cdot\frac{1}{4}}{(1-\frac{1}{4})^2} = \frac{\frac{3}{4}}{\frac{9}{16}} = \frac{4}{3}$

for ①: geometric series satisfied $\sum_{x=0}^{\infty} r^x = \frac{1}{1-r}$, $\Rightarrow \frac{d}{dr}\sum_{x=0}^{\infty} r^x = \frac{1}{(1-r)^2}$ $\Rightarrow \sum_{x=0}^{\infty} x\cdot r^{x-1} = \frac{1}{(1-r)^2}$

$\Rightarrow \sum_{x=0}^{\infty} x r^x = \frac{r}{(1-r)^2}$

(c.) $E(X) = \sum_{x=0}^{\infty}(\frac{x}{x+1}-\frac{x}{x+2})$, Let $a_x = (\frac{x}{x+1}-\frac{x}{x+2}) = \frac{x^2+2x-x^2-x}{(x+1)(x+2)} = \frac{x}{x^2+3x+2}$

$\lim_{x\to\infty}\left|\frac{a_x}{\frac{1}{x}}\right| = \lim_{x\to\infty}\frac{x^2}{x^2+3x+2} = \lim_{x\to\infty}\frac{1}{1+\frac{3}{x}+\frac{2}{x^2}} = 1$, $\because \sum_{x=1}^{\infty}\frac{1}{x}$ diverges, $\therefore \sum_{x=0}^{\infty} a_x$ diverges. $\Rightarrow E(X)$ doesn't exist.

T4. $E(X) = \sum_{x=0}^{\infty} x\cdot P(X=x) = \sum_{x=0}^{\infty}(\sum_{k=0}^{x-1} 1)\cdot P(X=x) = \sum_{x=0}^{\infty}\sum_{k=0}^{\infty} 1_{\{x>k\}} P(X=x) = \sum_{k=0}^{\infty}\sum_{x=0}^{\infty} 1_{\{x>k\}} P(X=x) = \sum_{k=0}^{\infty} P(X>k)$ □.

T5. $E(X) = \sum_{x=1}^{\infty}\frac{6}{\pi^2 x^2}\cdot x = \frac{6}{\pi^2}\cdot\sum_{x=1}^{\infty}\frac{1}{x}$, $\because \sum_{x=1}^{\infty}\frac{1}{x}$ diverges, $\therefore \frac{6}{\pi^2}\sum_{x=1}^{\infty}\frac{1}{x}$ diverges, $\frac{1}{x}>0$ for $x\in(1,+\infty) \Rightarrow E(X) = +\infty$ and thus, doesn't exist.